

# Engineering English

Instructor guide for advanced ESL learners working in engineering

**Audience: mechanical, electrical, civil, systems, industrial, test, quality, manufacturing, and field engineers, plus engineering managers and technical project leads**

**Focus:** An engineering English curriculum for requirements, design reviews, tradeoffs, testing, failure analysis, quality, safety factors, manufacturability, field issues, and technical disagreement.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

## Purpose and Course Logic

An engineering English curriculum for requirements, design reviews, tradeoffs, testing, failure analysis, quality, safety factors, manufacturability, field issues, and technical disagreement.

### Core language challenge

Advanced learners do not only need vocabulary. They need the ability to ask which standard applies, who owns the decision, what evidence is sufficient, what risk is being accepted, and how to disagree without sounding vague, defensive, or reckless.

Each module trains a realistic workplace pressure point with role-specific terms, decision language, pushback practice, and bounded decision activities learners can apply to their own work.

### Course objectives

- Use engineering terminology accurately in meetings, written updates, handoffs, escalations, reviews, and client or stakeholder conversations.
- Turn vague requests into specific questions about evidence, owner, deadline, constraint, risk, and decision rights.
- Push back on unsafe, unsupported, noncompliant, unrealistic, or poorly scoped proposals while preserving professional trust.
- Handle realistic dialogues from the field, including conflict, uncertainty, documentation gaps, customer or stakeholder pressure, and cross-functional disagreement.
- Select language that produces concise workplace outputs: briefing notes, escalation updates, meeting scripts, risk memos, decision records, and follow-up messages.

## Instructor Module Plans

### Module 1. Requirements and Constraints (90 minutes)

Clarify requirements before proposing technical solutions.

#### Learners should be able to

- Use these terms accurately: requirement, constraint, tolerance, tradeoff.
- Explain the workplace tension: Load, tolerance, material, manufacturability, regulatory, and cost constraints conflict.
- Respond professionally when a stakeholder says: Say engineering can make it work.
- Select the evidence, tradeoff, owner, and decision required for a requirements clarification matrix.

#### Customized scenario

##### Workplace pressure

A customer asks for a lighter design with no cost increase.

Say engineering can make it work.

Load, tolerance, material, manufacturability, regulatory, and cost constraints conflict.

#### Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.

- Decision artifact check: select the facts, caveat, owner, and next action that belong in a requirements clarification matrix.

## Module 2. Design Reviews and Technical Pushback (90 minutes)

Challenge assumptions using evidence rather than status.

### Learners should be able to

- Use these terms accurately: design review, design rationale, risk, verification.
- Explain the workplace tension: Design rationale, risk, analysis, and verification evidence should be reviewed.
- Respond professionally when a stakeholder says: Accept the senior view.
- Select the evidence, tradeoff, owner, and decision required for a design-review comment set.

### Customized scenario

#### Workplace pressure

A senior engineer prefers a design with limited test data.

Accept the senior view.

Design rationale, risk, analysis, and verification evidence should be reviewed.

### Classroom sequence

- Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
- Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
- Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
- Decision artifact check: select the facts, caveat, owner, and next action that belong in a design-review comment set.

## Module 3. Failure Modes and Reliability (90 minutes)

Explain failure risk, severity, and detection.

### Learners should be able to

- Use these terms accurately: failure mode, FMEA, reliability, duty cycle.
- Explain the workplace tension: Operating envelope, duty cycle, failure mode, and reliability target matter.
- Respond professionally when a stakeholder says: Treat it as an edge case.
- Select the evidence, tradeoff, owner, and decision required for a failure mode summary.

### Customized scenario

#### Workplace pressure

A component failure appears only under certain vibration conditions.

Treat it as an edge case.

Operating envelope, duty cycle, failure mode, and reliability target matter.

### Classroom sequence

- Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
- Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.

3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a failure mode summary.

## Module 4. Testing, Validation, and Data Interpretation (90 minutes)

Discuss test results without overgeneralizing.

### Learners should be able to

- Use these terms accurately: prototype, acceptance criteria, validation, sample size.
- Explain the workplace tension: Test conditions, sample size, acceptance criteria, and failure analysis are incomplete.
- Respond professionally when a stakeholder says: Claim the design is mostly validated.
- Select the evidence, tradeoff, owner, and decision required for a test-readiness update.

### Customized scenario

#### Workplace pressure

A prototype passes one test but fails under thermal cycling.

Claim the design is mostly validated.

Test conditions, sample size, acceptance criteria, and failure analysis are incomplete.

### Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a test-readiness update.

## Module 5. Manufacturability and Cost Engineering (90 minutes)

Connect design decisions to production and cost reality.

### Learners should be able to

- Use these terms accurately: DFM, tooling, yield, cycle time.
- Explain the workplace tension: Yield, tooling, cycle time, supplier capability, and cost must be balanced.
- Respond professionally when a stakeholder says: Approve it because performance is better.
- Select the evidence, tradeoff, owner, and decision required for a DFM tradeoff note.

### Customized scenario

#### Workplace pressure

A design change improves performance but complicates assembly.

Approve it because performance is better.

Yield, tooling, cycle time, supplier capability, and cost must be balanced.

### Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.

2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a DFM tradeoff note.

## Module 6. Safety Factors and Compliance (90 minutes)

Explain safety margin and regulatory implications clearly.

### Learners should be able to

- Use these terms accurately: safety factor, code compliance, margin, liability.
- Explain the workplace tension: Safety factor, code requirements, test evidence, and liability exposure need review.
- Respond professionally when a stakeholder says: Reduce it if simulations pass.
- Select the evidence, tradeoff, owner, and decision required for a safety margin escalation.

### Customized scenario

#### Workplace pressure

Leadership wants to reduce material thickness to save cost.

Reduce it if simulations pass.

Safety factor, code requirements, test evidence, and liability exposure need review.

### Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a safety margin escalation.

## Module 7. Field Issues and Customer Communication (90 minutes)

Communicate technical problems to customers without speculation.

### Learners should be able to

- Use these terms accurately: field failure, warranty, corrective action, installation condition.
- Explain the workplace tension: Evidence, installation conditions, warranty terms, and corrective action are not complete.
- Respond professionally when a stakeholder says: Tell the customer the part was misused.
- Select the evidence, tradeoff, owner, and decision required for a customer technical update.

### Customized scenario

#### Workplace pressure

A field failure affects a strategic account.

Tell the customer the part was misused.

Evidence, installation conditions, warranty terms, and corrective action are not complete.

## Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a customer technical update.

## Module 8. Systems Integration and Interface Control (90 minutes)

Manage cross-disciplinary dependencies.

### Learners should be able to

- Use these terms accurately: interface control, integration, regression test, configuration management.
- Explain the workplace tension: Interfaces, timing assumptions, version control, and regression testing need governance.
- Respond professionally when a stakeholder says: Ask teams to coordinate informally.
- Select the evidence, tradeoff, owner, and decision required for a interface-control update.

### Customized scenario

#### Workplace pressure

A software change affects hardware timing.

Ask teams to coordinate informally.

Interfaces, timing assumptions, version control, and regression testing need governance.

### Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a interface-control update.

## Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

### Requirements and Constraints

| Term        | Working meaning                                                                                                                                                                                                  | Collocations                                          | Contrast and workplace line                                                                                                                                         |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| requirement | In engineering, requirement is an approved rule or decision framework that sets what is permitted, required, or acceptable. Use it when teams need to clarify requirements before proposing technical solutions. | apply the requirement;<br>comply with the requirement | Contrast: requirement is an approved rule or threshold, not an informal preference.<br>Example: Please confirm which requirement applies before we commit the team. |

| Term       | Working meaning                                                                                                                                                                                               | Collocations                               | Contrast and workplace line                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| constraint | In engineering, constraint is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions. | clarify constraint;<br>document constraint | Contrast: constraint should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify constraint and the evidence that supports it. |
| tolerance  | In engineering, tolerance is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions.  | clarify tolerance;<br>document tolerance   | Contrast: tolerance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tolerance and the evidence that supports it.   |
| tradeoff   | In engineering, tradeoff is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to clarify requirements before proposing technical solutions.   | clarify tradeoff;<br>document tradeoff     | Contrast: tradeoff should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tradeoff and the evidence that supports it.     |

## Design Reviews and Technical Pushback

| Term             | Working meaning                                                                                                                                                                                                               | Collocations                                                    | Contrast and workplace line                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| design review    | In engineering, design review is a structured examination of evidence, process, or results used to reach a defensible conclusion. Use it when teams need to challenge assumptions using evidence rather than status.          | conduct a design review;<br>document the design review          | Contrast: design review is a structured examination of evidence; it is not a conclusion reached before the evidence is checked. Example: The team will conduct a design review before it recommends a corrective action.    |
| design rationale | In engineering, design rationale is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to challenge assumptions using evidence rather than status. | track design rationale;<br>explain a change in design rationale | Contrast: design rationale measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track design rationale by segment and explain what changed.                              |
| risk             | In engineering, risk is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to challenge assumptions using evidence rather than status.                | assess risk; mitigate risk                                      | Contrast: risk identifies a possible or current exposure; it is not the same as accepting that exposure. Example: We need to assess risk, name the owner, and agree on the trigger for escalation.                          |
| verification     | In engineering, verification is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to challenge assumptions using evidence rather than status.        | clarify verification;<br>document verification                  | Contrast: verification should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify verification and the evidence that supports it. |

## Failure Modes and Reliability

| Term         | Working meaning                                                                                                                                                                                              | Collocations                                   | Contrast and workplace line                                                                                                                                                                                                 |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| failure mode | In engineering, failure mode is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to explain failure risk, severity, and detection. | clarify failure mode;<br>document failure mode | Contrast: failure mode should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify failure mode and the evidence that supports it. |

| Term        | Working meaning                                                                                                                                                                                                                | Collocations                                 | Contrast and workplace line                                                                                                                                                                                               |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FMEA        | Failure mode and effects analysis, a structured method for identifying how a process or product can fail and prioritizing controls.                                                                                            | clarify FMEA;<br>document FMEA               | Contrast: FMEA should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify FMEA and the evidence that supports it.               |
| reliability | In engineering, reliability is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain failure risk, severity, and detection. | clarify reliability;<br>document reliability | Contrast: reliability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify reliability and the evidence that supports it. |
| duty cycle  | In engineering, duty cycle is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain failure risk, severity, and detection.                              | clarify duty cycle;<br>document duty cycle   | Contrast: duty cycle should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify duty cycle and the evidence that supports it.   |

## Testing, Validation, and Data Interpretation

| Term                | Working meaning                                                                                                                                                                                              | Collocations                                                          | Contrast and workplace line                                                                                                                                                                                               |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prototype           | In engineering, prototype is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss test results without overgeneralizing.             | clarify prototype;<br>document prototype                              | Contrast: prototype should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify prototype and the evidence that supports it.     |
| acceptance criteria | In engineering, acceptance criteria is an approved rule or decision framework that sets what is permitted, required, or acceptable. Use it when teams need to discuss test results without overgeneralizing. | apply the acceptance criteria;<br>comply with the acceptance criteria | Contrast: acceptance criteria is an approved rule or threshold, not an informal preference. Example: Please confirm which acceptance criteria applies before we commit the team.                                          |
| validation          | In engineering, validation is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to discuss test results without overgeneralizing.   | clarify validation;<br>document validation                            | Contrast: validation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify validation and the evidence that supports it.   |
| sample size         | In engineering, sample size is a defined information asset or analytical construct used to measure, explain, or predict an outcome. Use it when teams need to discuss test results without overgeneralizing. | clarify sample size;<br>document sample size                          | Contrast: sample size should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify sample size and the evidence that supports it. |

## Manufacturability and Cost Engineering

| Term | Working meaning                                                                                                                                                                                      | Collocations              | Contrast and workplace line                                                                                                                                                                               |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DFM  | In engineering, DFM is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality. | clarify DFM; document DFM | Contrast: DFM should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify DFM and the evidence that supports it. |



| Term       | Working meaning                                                                                                                                                                                                    | Collocations                               | Contrast and workplace line                                                                                                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tooling    | In engineering, tooling is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality.           | clarify tooling;<br>document tooling       | Contrast: tooling should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tooling and the evidence that supports it.       |
| yield      | In engineering, yield is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to connect design decisions to production and cost reality. | improve yield; analyze yield loss          | Contrast: Yield measures conforming output, not the root cause of a process change. Example: The fab is analyzing yield loss by tool, layer, and time window.                                                           |
| cycle time | In engineering, cycle time is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to connect design decisions to production and cost reality.        | clarify cycle time;<br>document cycle time | Contrast: cycle time should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify cycle time and the evidence that supports it. |

## Safety Factors and Compliance

| Term            | Working meaning                                                                                                                                                                                                                                | Collocations                                      | Contrast and workplace line                                                                                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safety factor   | In engineering, safety factor is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain safety margin and regulatory implications clearly.   | clarify safety factor;<br>document safety factor  | Contrast: safety factor should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify safety factor and the evidence that supports it.     |
| code compliance | In engineering, code compliance is a performance or governance dimension that must be protected through defined evidence, thresholds, and accountability. Use it when teams need to explain safety margin and regulatory implications clearly. | clarify code compliance; document code compliance | Contrast: code compliance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify code compliance and the evidence that supports it. |
| margin          | In engineering, margin is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to explain safety margin and regulatory implications clearly.                          | track margin; explain a change in margin          | Contrast: margin measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track margin by segment and explain what changed.                                                        |
| liability       | In engineering, liability is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain safety margin and regulatory implications clearly.                                   | clarify liability;<br>document liability          | Contrast: liability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify liability and the evidence that supports it.             |

## Field Issues and Customer Communication

| Term                   | Working meaning                                                                                                                                                                                                                 | Collocations                                                       | Contrast and workplace line                                                                                                                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| field failure          | In engineering, field failure is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to communicate technical problems to customers without speculation. | clarify field failure;<br>document field failure                   | Contrast: field failure should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify field failure and the evidence that supports it.                   |
| warranty               | In engineering, warranty is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to communicate technical problems to customers without speculation.               | clarify warranty;<br>document warranty                             | Contrast: warranty should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify warranty and the evidence that supports it.                             |
| corrective action      | Action taken to fix a current problem and prevent recurrence.                                                                                                                                                                   | clarify corrective action;<br>document corrective action           | Contrast: corrective action should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify corrective action and the evidence that supports it.           |
| installation condition | In engineering, installation condition is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to communicate technical problems to customers without speculation. | clarify installation condition;<br>document installation condition | Contrast: installation condition should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify installation condition and the evidence that supports it. |

## Systems Integration and Interface Control

| Term                     | Working meaning                                                                                                                                                                                                      | Collocations                                                                    | Contrast and workplace line                                                                                                                                                                                                           |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| interface control        | In engineering, interface control is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to manage cross-disciplinary dependencies.           | clarify interface control;<br>document interface control                        | Contrast: interface control should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify interface control and the evidence that supports it. |
| integration              | In engineering, integration is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to manage cross-disciplinary dependencies.              | track integration;<br>explain a change in integration                           | Contrast: integration measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track integration by segment and explain what changed.                                                  |
| regression test          | In engineering, regression test is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to manage cross-disciplinary dependencies.                      | clarify regression test;<br>document regression test                            | Contrast: regression test should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify regression test and the evidence that supports it.     |
| configuration management | In engineering, configuration management is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to manage cross-disciplinary dependencies. | track configuration management;<br>explain a change in configuration management | Contrast: configuration management measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track configuration management by segment and explain what changed.                        |

## Industry-Specific Meeting Moves

| Situation                                    | Useful language                                                                                                                                                                                                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirements and Constraints                 | Before we commit, I want to confirm requirement, constraint, the owner, and the evidence behind the decision. If load, tolerance, material, manufacturability, regulatory, and cost constraints conflict., I recommend we document the risk and agree on the next step.        |
| Design Reviews and Technical Pushback        | Before we commit, I want to confirm design review, design rationale, the owner, and the evidence behind the decision. If design rationale, risk, analysis, and verification evidence should be reviewed., I recommend we document the risk and agree on the next step.         |
| Failure Modes and Reliability                | Before we commit, I want to confirm failure mode, FMEA, the owner, and the evidence behind the decision. If operating envelope, duty cycle, failure mode, and reliability target matter., I recommend we document the risk and agree on the next step.                         |
| Testing, Validation, and Data Interpretation | Before we commit, I want to confirm prototype, acceptance criteria, the owner, and the evidence behind the decision. If test conditions, sample size, acceptance criteria, and failure analysis are incomplete., I recommend we document the risk and agree on the next step.  |
| Manufacturability and Cost Engineering       | Before we commit, I want to confirm DFM, tooling, the owner, and the evidence behind the decision. If yield, tooling, cycle time, supplier capability, and cost must be balanced., I recommend we document the risk and agree on the next step.                                |
| Safety Factors and Compliance                | Before we commit, I want to confirm safety factor, code compliance, the owner, and the evidence behind the decision. If safety factor, code requirements, test evidence, and liability exposure need review., I recommend we document the risk and agree on the next step.     |
| Field Issues and Customer Communication      | Before we commit, I want to confirm field failure, warranty, the owner, and the evidence behind the decision. If evidence, installation conditions, warranty terms, and corrective action are not complete., I recommend we document the risk and agree on the next step.      |
| Systems Integration and Interface Control    | Before we commit, I want to confirm interface control, integration, the owner, and the evidence behind the decision. If interfaces, timing assumptions, version control, and regression testing need governance., I recommend we document the risk and agree on the next step. |

## High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.
- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

## Assessment and Coaching

### Performance rubric

| Skill                  | Developing                                           | Proficient                                               | Strong                                                                                    |
|------------------------|------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Terminology            | Recognizes terms but uses them loosely.              | Uses field terms accurately in context.                  | Defines terms, connects them to evidence, and explains decision impact.                   |
| Pushback               | Disagrees vaguely or avoids disagreement.            | Names concern with evidence and next step.               | Balances urgency, relationship, risk, owner, and decision rights.                         |
| Scenario judgment      | Focuses on one stakeholder's preference.             | Identifies constraint, risk, and process.                | Guides the group toward a documented, realistic decision.                                 |
| Decision communication | Selects language without linking it to the decision. | Chooses language that names facts, owner, and next step. | Distinguishes evidence, tradeoff, authority, and escalation in a decision-ready response. |

## Source orientation

- Engineering standards and applicable codes.
- Company design-control, test, and quality procedures.
- Customer specifications and contractual requirements.
- The learner's own company policies, SOPs, contracts, systems, templates, and approved communication standards.